

Chocolate Desert Millipede Care Guide

Orthoporus ornatus

Red Rocks Exotic Reptiles · Reviewed October 2026 · Version 2.0

Desert origin does not mean dry bedding. Moist deep soil is essential for hydration, feeding and underground molts; combine it with a drier, ventilated surface.

Species Profile

Common name: Chocolate Desert Millipede

Scientific name: *Orthoporus ornatus*

Adult size: About 4-8 in (10-20 cm) body length; locality varies

Captive lifespan: Reliable captive average is not established; age of wild-caught adults is usually unknown

Native range: Southwestern United States and northern Mexico

Habitat and activity: Burrowing detritivore; seasonal surface activity in semi-arid habitat

Diet: Decaying hardwood leaves and wood, with small vegetable portions

Care difficulty: Beginner to Intermediate

Shopping List

- Secure ventilated enclosure with the floor area and substrate depth described under Housing.
- Pesticide-free decaying hardwood leaf litter, soft white-rotted hardwood and suitable soil; low cork cover.
- Sprayer, digital thermometer, and small portions of supplementary vegetables.
- Optional springtails; a small calcium source can be offered separately.

Housing

A footprint of about 24 × 12 in (61 × 30 cm) is a useful starting point for a few adults. Provide substrate at least as deep as the animal is long where practical; start with 6-8 in (15-20 cm) or more. More space and food are needed as the colony grows.

Lighting & UVB

Provide an ordinary day/night cycle with shaded retreats. Specialized UVB is not an established requirement for these millipedes. Avoid direct sunlight and lamps that heat or dry the substrate.

Heating & Temperature

A useful daytime range is 72-80°F (22-27°C), with cooler sheltered areas and a modest night drop toward 68-75°F (20-24°C). Ordinary room conditions often suffice; measure the actual enclosure.

Keep the enclosure away from direct sun, radiators and hot equipment. Do not use bottom heat to dry burrowing substrate. If the room is too cold, stabilizing the room is usually easier to control than heating a small enclosure directly.

Humidity & Ventilation

Dry surface areas with continuously moist deep refuges; never dry all substrate. Check moisture below the surface with minimal disturbance at an edge. Air humidity alone cannot tell you whether a buried animal has access to water.

Water one portion of the substrate as needed so moist deep refuges remain available. Keep drainage and ventilation effective. Do not dry the entire enclosure or flood it to create humidity.

Water & Hydration

Hydration comes primarily from moist substrate and moisture-rich food. A large open water dish is unnecessary and can drown small offspring. Maintain moisture in the deeper refuge without digging through occupied molt chambers.

Use clean water and equipment free of pesticide or detergent residue. Small produce portions support hydration but cannot replace moist substrate or the main detritus diet.

Substrate

The substrate is both habitat and food. Provide at least 6 inches for adults, and deeper is better when enclosure height allows. Desert millipedes use the lower layers for resting, molting, and escaping dry surface conditions.

A practical mix is pesticide- and fertilizer-free organic topsoil with roughly 20-30% washed play sand, plus generous amounts of decaying hardwood leaf litter and soft white-rotted hardwood. Mix some edible leaves and wood throughout the soil and keep additional leaf litter on the surface.

Avoid: cedar, pine, aromatic softwoods, chemically treated wood, fertilized potting soil, pesticide-exposed yard debris, and using plain coconut fiber as the main substrate. A sandy-looking enclosure still needs edible hardwood material mixed through the soil.

Food & Feeding

Most of a chocolate desert millipede's diet should come from decomposing leaves, decayed hardwood, and other dead plant material already inside the enclosure. Keep these foods available continuously and replenish them as they break down.

Supplement the substrate with small portions of squash, sweet potato, carrot, cucumber, zucchini, mushroom, or leafy greens 1-3 times per week. Fruit such as banana or melon can be offered occasionally. Remove soft leftovers before they spoil.

Optional protein: A very small pinch of fish flakes, shrimp pellets, nutritional yeast, or a balanced isopod/millipede food can be offered about once a week. This is supplemental - not a replacement for leaf litter and decaying wood.

Supplements

A small separate piece of cuttlebone or a little plain calcium carbonate can be offered. The optimum supplemental calcium dose is not established, and a calcium source cannot compensate for poor edible substrate. Routine reptile multivitamin or D3 dusting is not recommended.

Decor & Enrichment

Provide cork bark, dry hardwood pieces, leaf litter, and several sturdy branches or pieces of bark.

These create shaded resting areas and additional surfaces for grazing and climbing. Keep the decor stable so a burrowing millipede cannot undermine a heavy object and become trapped beneath it.

Handling & Behavior

Handle briefly, with clean hands, close to the enclosure or a soft surface. Support the body and let the animal walk; never pull an animal out of its coil or excavate it for handling.

Defensive secretions can irritate skin and eyes or stain skin. Wash hands afterward, keep them away from eyes and mouth, and avoid contact for anyone with sensitivity. Falls can cause serious injury.

Cleaning & Routine Maintenance

Check moisture, temperature, fresh-food spoilage and the supply of edible leaves/wood regularly. Remove spoiled extras while preserving the established detritus substrate and buried refuges.

Replenish decomposing leaves and wood as consumed. Do not perform a deep full substrate change while animals may be molting underground. If contamination demands intervention, plan the transfer carefully and preserve safe moist chambers where possible.

Cohabitation

Same-species colonies are appropriate when edible substrate, moisture and space are sufficient. Mixed-sex colonies may reproduce. Avoid mixing species with different moisture needs or adding predatory/competitive cleanup animals without understanding their effects on eggs and molts.

Shedding, Molting & Seasonal Changes

Buried animals may remain out of sight for weeks or longer. Molting is an extremely vulnerable period: leave chambers undisturbed and maintain safe deep moisture and temperature. Do not dig up an animal simply to confirm it is alive.

Surface activity can vary with environmental conditions. Reduced activity is not a reason to dry all soil, force-feed or repeatedly handle the animal. Review conditions if several animals show abnormal behavior or decline.

Reproduction

Mixed-sex colonies may lay eggs in the substrate and produce tiny offspring. Keep edible substrate and appropriate deep moisture available, use fine ventilation gaps and avoid deep stirring that destroys egg or molt chambers. Population growth may require additional space and food.

Common Health Concerns

- Dehydration or unsuitable edible substrate: review deep moisture and the ongoing leaf/wood supply; fresh fruit alone will not correct poor habitat.
- Failed molts or injury: do not disturb underground animals; avoid falls and hazardous sharp decor.

- Contamination: pesticide residues, treated wood and spoiled supplemental food can harm a colony. Persistent decline deserves advice from an experienced millipede keeper or an exotic veterinarian familiar with invertebrates.

Sources & Review Information

Reviewed October 2026 · RRE version 2.0. Care targets are practical starting recommendations informed by natural history, published work and captive experience. For less-studied species, enclosure dimensions, feeding frequencies, UVI ranges and lifespan estimates are not experimentally validated universal thresholds. Adjust for the individual, locality, life stage and veterinary advice.

Water relations in *Orthoporus ornatus* (1972). Moist-substrate water uptake and desiccation resistance. [https://doi.org/10.1016/0300-9629\(72\)90131-4](https://doi.org/10.1016/0300-9629(72)90131-4)

Crawford (1978). Seasonal water balance in *Orthoporus ornatus*. <https://doi.org/10.2307/1938551>

RED ROCKS
EXOTIC REPTILES
—★ ★ ★ ★—